TECHNICAL REQUIREMENTS DOCUMENT (TRD)

Mehmoodah Academy — School Operating System (V1)

Internal School Platform

Version: 2.0
Status: Ready for Development

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1. SYSTEM OVERVIEW

Mehmoodah Academy School Operating System is a web application designed to manage internal school operations.

Users:

- Admin
- Teachers
- Students

Scope:

- Attendance
- Student Records
- Results
- Homework
- Timetable
- Reports

Architecture: Frontend → Backend → Database → Storage

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2. SYSTEM ARCHITECTURE

CLIENT

Flutter Web

↓

API LAYER

Business Logic

↓

DATABASE

PostgreSQL

↓

FILE STORAGE

Cloud Storage

↓

MONITORING

Logs + Analytics

Architecture Type: Modular Monolith

Future: Multi-school ready

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3. TECH STACK

Frontend: Flutter Web

State Management: Riverpod

Backend: Supabase

Database: PostgreSQL

Authentication: Supabase Auth

Storage: Supabase Storage

Hosting: Vercel

Notifications: Firebase

Version Control: GitHub

4. APPLICATION STRUCTURE

frontend/

lib/

core/

modules/

auth/

dashboard/

students/

teachers/

attendance/

results/

assignments/

timetable/

reports/

profile/

shared/

widgets/

services/

models/

routes/

backend/

database/

policies/

functions/

storage/

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5. ROLE ACCESS CONTROL

ADMIN

Access: All Modules

Permissions: Create Read Update Delete

TEACHER

Access: Assigned Classes

Permissions: Read Update

Restricted: Delete

STUDENT

Access: Own Academic Records

Permissions: Read Only

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6. DATABASE DESIGN

TABLE: users

id email role created_at

TABLE: teachers

id user_id name phone subject status

TABLE: students

id student_id name class_id roll_no guardian_name

TABLE: classes

id name section teacher_id

TABLE: attendance

id student_id class_id teacher_id date status

TABLE: exams

id name class_id term

TABLE: results

id exam_id student_id marks grade

TABLE: assignments

id teacher_id class_id title due_date

TABLE: submissions

id assignment_id student_id status

TABLE: announcements

id title body published_at

TABLE: timetable

id class_id teacher_id day

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7. AUTH FLOW

User Login

↓

Validate Credentials

↓

Issue Session

↓

Load Role

↓

Redirect Dashboard

Security: JWT

Session: 7 Days

Refresh: Automatic

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8. TEACHER CREATION FLOW

Admin

↓

Open Teachers

↓

Add Teacher

↓

Create Auth User

↓

Insert Teacher Record

↓

Assign Role

↓

Activate Account

Validation: Unique Email

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9. STUDENT CREATION FLOW

Admin

↓

Open Students

↓

Create Student

↓

Generate Student ID

↓

Assign Class

↓

Create Student Login

↓

Save

Validation: Unique Student ID

Bulk Upload: CSV Supported

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10. ATTENDANCE FLOW

Teacher

↓

Select Class

↓

Load Students

↓

Mark Status

↓

Submit

↓

Store Records

Rules:

One attendance/day

Prevent duplicates

Statuses:

Present

Absent

Late

Leave

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11. RESULT FLOW

Teacher

↓

Create Exam

↓

Input Marks

↓

Calculate %

↓

Generate Grade

↓

Publish

Student:

Read Only

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12. ASSIGNMENT FLOW

Teacher

↓

Create Assignment

↓

Upload File

↓

Publish

Student

↓

View

↓

Submit

Teacher

↓

Review

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13. STORAGE

Store:

Photos

Homework

Documents

Reports

Structure:

students/

teachers/

assignments/

reports/

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14. API ENDPOINTS

AUTH

POST /login

POST /logout

POST /reset

STUDENTS

GET /students

POST /students

PUT /students/{id}

DELETE /students/{id}

TEACHERS

GET /teachers

POST /teachers

PUT /teachers/{id}

ATTENDANCE

GET /attendance

POST /attendance

RESULTS

GET /results

POST /results

ASSIGNMENTS

GET /assignments

POST /assignments

REPORTS

GET /reports

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15. PERFORMANCE

Dashboard: <2 sec

Attendance: <1 sec

Reports: <5 sec

Search: <500 ms

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16. SECURITY

RBAC

HTTPS

Input Validation

Audit Logs

Rate Limits

Row Level Security

Password Hashing

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17. ERROR HANDLING

400 Validation

401 Unauthorized

403 Forbidden

404 Missing

500 Server

UI: Readable Errors

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18. DEPLOYMENT

DEV

↓

STAGING

↓

PRODUCTION

Pipeline:

GitHub

↓

Build

↓

Deploy

↓

Monitor

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19. BACKUP

Database: Daily

Storage: Weekly

Retention: 30 Days

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20. DEVELOPMENT ORDER

Phase 1 Auth

Phase 2 Students

Phase 3 Teachers

Phase 4 Attendance

Phase 5 Results

Phase 6 Assignments

Phase 7 Reports

Phase 8 Launch

END OF TRD